

**Answer ALL questions.**

**Write your answers in the spaces provided.**

- 1** The stigma of a flower secretes a solution of sucrose. Pollen grains germinate in this solution to produce pollen tubes.

The effect of sucrose concentration on the germination of pollen grains was investigated.

Pollen grains from a single flower were scraped into the cavity of a slide containing sucrose solution.

After one hour, the slide was viewed through the high power of a microscope.

The number of germinating pollen grains was counted together with the total number of grains in the field of view.

This procedure was repeated for each sucrose solution, using pollen grains from the same flower.

The percentage germination of pollen grains was calculated.

- (a) (i) State the independent variable in this investigation.

(1)

- (ii) State **one** abiotic (environmental) variable that should be controlled in this investigation.

(1)

- (iii) Explain why the results might have been affected if the variable you named in (a)(ii) had not been controlled.

(3)

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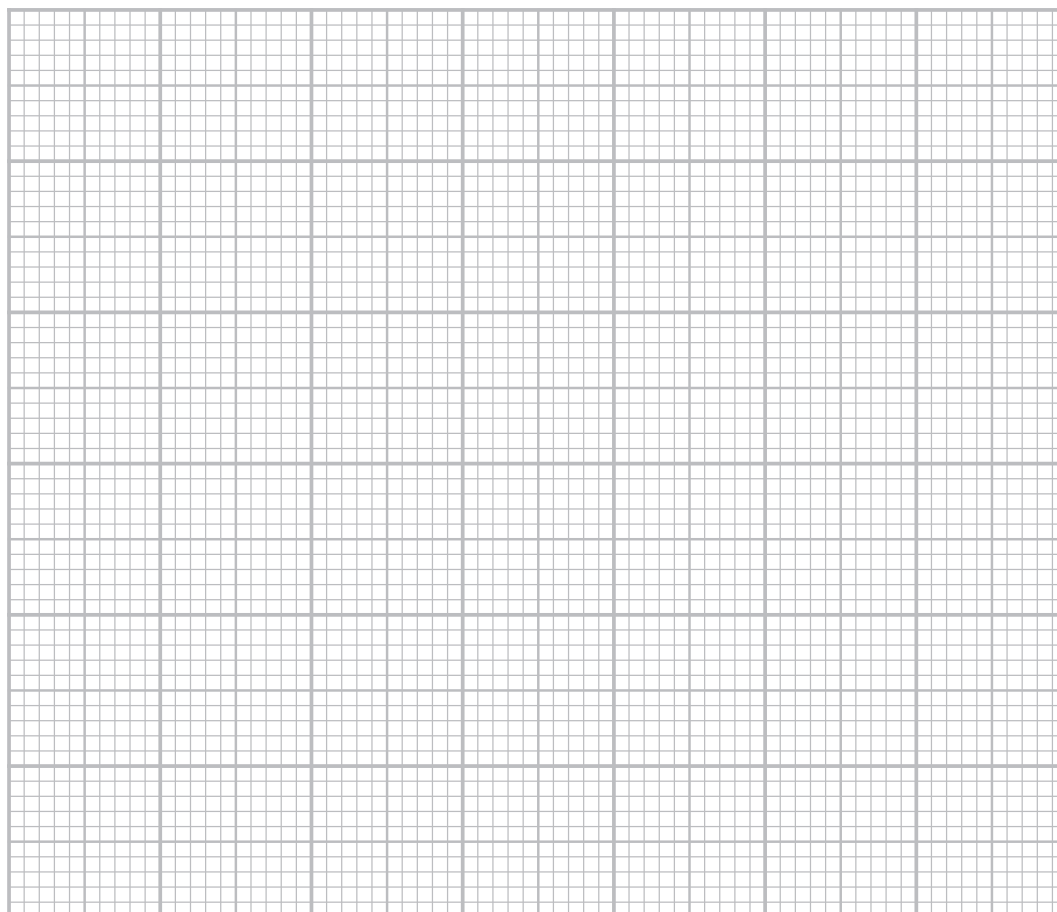


(b) The table shows the results of this investigation.

| Concentration of sucrose solution<br>/ g dm <sup>-3</sup> | Percentage germination of<br>pollen grains (%) |
|-----------------------------------------------------------|------------------------------------------------|
| 0                                                         | 10                                             |
| 20                                                        | 39                                             |
| 40                                                        | 73                                             |
| 60                                                        | 50                                             |
| 80                                                        | 41                                             |
| 100                                                       | 0                                              |
| 120                                                       | 0                                              |

- (i) Plot a suitable graph to show the results of this investigation.  
Join the points with straight lines.

(4)



P 6 1 4 7 1 A 0 3 1 6

(ii) These results suggest that the optimum concentration of sucrose for pollen grain germination is 40 g dm<sup>-3</sup>.

Explain how the method could be modified to find the actual optimum concentration of sucrose.

(2)

(c) The table shows the results of an investigation into the effect of boric acid concentration, in a sucrose solution, on the growth of pollen tubes.

| Boric acid concentration / parts per million (ppm) | Pollen tube length / $\mu\text{m}$ |
|----------------------------------------------------|------------------------------------|
| 25                                                 | 78                                 |
| 50                                                 | 126                                |
| 100                                                | 166                                |
| 200                                                | 134                                |
| 300                                                | 112                                |
| 400                                                | 90                                 |

(i) Calculate the percentage increase in pollen tube length at 100 ppm compared with that at 25 ppm.

(2)

Answer .....%



- (ii) You are provided with a solution containing  $500 \text{ g dm}^{-3}$  of sucrose.

Devise a procedure to investigate the effect of different concentrations of sucrose on the rate of growth of pollen tubes, using this sucrose solution.

(5)

(Total for Question 1 = 18 marks)



P 6 1 4 7 1 A 0 5 1 6